



7782 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Kevin Luhman (PI)	The Pennsylvania State University
Dr. Catarina Alves de Oliveira (CoI) (ESA Member)	ESA, European Space Astronomy Centre
Dr. Lynne A. Hillenbrand (CoI)	California Institute of Technology
Dr. Catherine Espaillat (CoI)	Boston University
Dr. Nuria Calvet (CoI)	University of Michigan

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec MOS placeholders				
	1	o1a, o1b, o1c, o1d	NIRSpec MultiObject Spectroscopy	(2) o1
	2	o2a, o2b, o2c	NIRSpec MultiObject Spectroscopy	(3) o2
	3	o3a, o3b, o3c	NIRSpec MultiObject Spectroscopy	(6) o3
	4	o4c, o4d	NIRSpec MultiObject Spectroscopy	(4) o4
	5	o5a, o5b	NIRSpec MultiObject Spectroscopy	(7) o5
	6	o6a, o6b, o6c, o6d, o6e	NIRSpec MultiObject Spectroscopy	(5) o6
	7	o7a, o7b, o7c, o7d, o7e	NIRSpec MultiObject Spectroscopy	(9) o7
	8	o8a, o8b, o8c, o8d	NIRSpec MultiObject Spectroscopy	(8) o8
	16	o6a, o6b, o6c, o6d, o6e	NIRSpec MultiObject Spectroscopy	(5) o6
	17	o17a, o17c, o17d2, o17b2	NIRSpec MultiObject Spectroscopy	(10) o17
NIRCam Persistence Mitigation				

JWST Proposal 7782 (Created: Thursday, March 12, 2026, 5:01:35PM Eastern Standard Time) - Overview

Folder	Observation	Label	Observing Template	Science Target
	11	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	12	Residual Persistence Check	NIRCcam Dark	NONE
	21	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	22	Residual Persistence Check	NIRCcam Dark	NONE
	31	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	32	Residual Persistence Check	NIRCcam Dark	NONE
	41	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	42	Residual Persistence Check	NIRCcam Dark	NONE
	51	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	52	Residual Persistence Check	NIRCcam Dark	NONE
	61	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	62	Residual Persistence Check	NIRCcam Dark	NONE
	63	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	64	Residual Persistence Check	NIRCcam Dark	NONE
	71	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	72	Residual Persistence Check	NIRCcam Dark	NONE
	81	Move NIRCcam PWs to FLAT	NIRCcam Dark	NONE
	82	Residual Persistence Check	NIRCcam Dark	NONE

ABSTRACT

The Orion Nebula Cluster (ONC) is the richest and densest nearby star-forming cluster. As such, it is uniquely suited for imaging and multi-object spectroscopy with JWST. During Cycle 1, brief observations with NIRSpec provided spectra for 22 brown dwarf candidates identified with HST/WFC3, 21 of which were confirmed. Those targets included two of the coolest and least massive known proplyds and a strong candidate for a protostellar brown dwarf. Meanwhile, NIRCам images were obtained for a 11'x7.5' field in the center of the ONC, which have been used to identify >200 brown dwarf candidates that lack spectroscopy, the faintest of which could have masses of 1-2 Jupiter masses. This sample includes dozens of candidates for protostars and proplyds at substellar masses.

We propose to use NIRSpec to obtain PRISM spectra (1-5 μm) of most of the brown dwarf candidates (~200) from the NIRCам images of the ONC. We will use these spectra to confirm the youth and cool nature of the candidates, search for ice absorption features from protostars, and search for strong emission lines from proplyds. Through these observations, we seek to significantly increase the numbers of spectroscopically confirmed protostellar brown dwarfs and brown dwarf proplyds (beyond the current values of ~1 and ~2-3) and obtain the best available measurement of the mass function and minimum mass of brown dwarfs, all of which are important for testing theories for the formation of brown dwarfs. Our survey will also make it possible to investigate the origin of the abundant unidentified hydrocarbon that has been detected by NIRSpec in planetary mass objects in the Perseus star-forming region.

OBSERVING DESCRIPTION

The APT file contains placeholders for 24 MSA configurations for NIRSpec that are distributed across 8 MPT observing plans/visits. The final configurations will be submitted after aperture position angles are assigned by the observatory. Each placeholder configuration uses the PRISM disperser, NRSRAPID readout pattern, 13 groups, 9 integrations, and 3 dithers, corresponding to a total exposure time of 4070 s. The numbers of groups and integrations may be adjusted to be appropriate for the range of magnitudes of the targets in each final configuration. During each NIRSpec exposure, coordinated parallel images with NIRCам will be obtained with the SHALLOW2 readout pattern, three groups, and two integrations through a pair of filters from among F140M, F162M, and F182 in the short wavelength channel and F360M and F444W in the long wavelength channel.

Proposal 7782 - Targets - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	mpt	RA: 05 35 15.2514 (83.8135475d) Dec: -05 23 5.87 (-5.38496d) Equinox: J2000		
	Comments: Description=[]				
	(2)	o1	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(3)	o2	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(4)	o4	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(5)	o6	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(6)	o3	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(7)	o5	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(8)	o8	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				
	(9)	o7	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000		
	Comments: Description=[]				

Proposal 7782 - Targets - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects			
	(10)	o17	RA: 05 35 15.6480 (83.8152000d)
			Dec: -05 23 8.17 (-5.38560d)
			Equinox: J2000
			Comments: Description=

Proposal 7782 - Observation 1 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 1: o1a, o1b, o1c, o1d											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCam Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	o1	RA: 05 35 15.6480 (83.8152000d)									
			Dec: -05 23 8.17 (-5.38560d)									
			Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCam Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p1 (466 sources)											
	Filler Candidate List: f1 (69 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Spectral Overlap Threshold: 1.5												
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	126	83.849490	-5.361986	22.029	1	263	83.852549	-5.367422	21.518		
	1	157	83.870026	-5.339004	21.075	1	746	83.886881	-5.381528	21.475		
	1	169	83.845537	-5.345231	21.104	1	753	83.881436	-5.328412	21.808		
	1	193	83.859952	-5.330830	22.058	1	763	83.880127	-5.327328	21.802		
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 1 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	2 (PRISM/CLEAR)	c1 : o1d	3 Shutter Slitlet	83.875174583333 32 Degrees - 5.353288055555 38 Degrees	217.01578534213 021			3	6	1030.73
	2	1 (PRISM/CLEAR)	c1 : o1b	3 Shutter Slitlet	83.871013749999 99 Degrees - 5.351155555555 79 Degrees	217.01616873443 23			3	15	3382.083
	3	1 (PRISM/CLEAR)	c1 : o1a	3 Shutter Slitlet	83.869643333333 33 Degrees - 5.361242222222 45 Degrees	217.01631765990 345			3	15	3382.083
	4	1 (PRISM/CLEAR)	c1 : o1c	3 Shutter Slitlet	83.864096666666 67 Degrees - 5.356952222222 19 Degrees	217.01682592168 302			3	15	3382.083
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	4	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 217.0214 to 217.0214 Degrees (V3 78.44688 to 78.44688)										
	Sequence Observations 1, 11, Non-interruptible										

Proposal 7782 - Observation 2 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 2: o2a, o2b, o2c											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(o2a, o2b, o2c (Obs 2)) Warning (Form): Config c1 : o2a (#3) has 1 primary slits affected by failed closed shutters.											
	(Visit 2:1) Warning (Form): Data Excess over lower threshold											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	o2	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCам Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p2 (450 sources)											
	Filler Candidate List: f2 (85 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Spectral Overlap Threshold: 1.5												
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	55	83.829238	-5.439176	21.290	1	792	83.865210	-5.448893	22.692		
	1	556	83.879932	-5.428075	20.868	1	797	83.863868	-5.443883	21.711		
	1	757	83.885085	-5.419976	22.697	1	801	83.856926	-5.439987	22.682		
	1	767	83.877455	-5.435321	21.833	1	838	83.833060	-5.438027	21.387		
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 2 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	2 (PRISM/CLEAR)	c1 : o2b	3 Shutter Slitlet	83.854407958333 33 Degrees - 5.428252222222 27 Degrees	218.82615946656 486			3	6	1352.833
	2	1 (PRISM/CLEAR)	c1 : o2c	3 Shutter Slitlet	83.855680416666 67 Degrees - 5.426266111111 9 Degrees	218.82603681623 524			3	24	5411.332
	3	1 (PRISM/CLEAR)	c1 : o2a	3 Shutter Slitlet	83.869188749999 99 Degrees - 5.428125277777 81 Degrees	218.82477877666 244			3	24	5411.332
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 218.8297 to 218.8297 Degrees (V3 80.25517 to 80.25517)										
	Sequence Observations 2, 21, Non-interruptible										

Proposal 7782 - Observation 3 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 3: o3a, o3b, o3c											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	o3	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p3 (394 sources)											
	Filler Candidate List: f3 (141 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Spectral Overlap Threshold: 1.5												
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	31	83.725684	-5.425243	21.782	1	686	83.730841	-5.431368	21.457		
	1	48	83.730835	-5.441074	20.669	1	928	83.773343	-5.410945	21.265		
	1	493	83.756498	-5.408433	20.975	1	929	83.773494	-5.415577	21.967		
	1	682	83.734614	-5.440304	21.089	1	974	83.739099	-5.436786	22.306		
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 3 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	2 (PRISM/CLEAR)	c1 : o3c	3 Shutter Slitlet	83.754917083333 32 Degrees - 5.426521666666 45 Degrees	214.29108596575 125			3	6	1352.833
	2	1 (PRISM/CLEAR)	c1 : o3a	3 Shutter Slitlet	83.750163333333 33 Degrees - 5.431525000000 22 Degrees	214.29154066519 004			3	24	5411.332
	3	1 (PRISM/CLEAR)	c1 : o3b	3 Shutter Slitlet	83.752888749999 98 Degrees - 5.440223333333 36 Degrees	214.29130165886 943			3	24	5411.332
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 214.2854 to 214.2854 Degrees (V3 75.7108 to 75.7108)										
	Sequence Observations 3, 31, Non-interruptible										

Proposal 7782 - Observation 4 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 4: o4c, o4d											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCam Imaging											
Diagnostics	(o4c, o4d (Obs 4)) Warning (Form): Config c1 : o4d (#2) has 1 filler slits affected by failed closed shutters.											
	(o4c, o4d (Obs 4)) Warning (Form): Config c1 : o4d (#2) has 1 primary slits affected by failed closed shutters.											
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	o4	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCam Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p4 (438 sources)											
	Filler Candidate List: f4 (97 sources)											
	Spectral Overlap Map: jwst-nirspec-prism											
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	37	83.810642	-5.441475	22.127	1	817	83.849385	-5.428558	20.874		
	1	524	83.824453	-5.439373	22.391	1	828	83.842090	-5.447838	22.204		
	1	800	83.856929	-5.428768	20.850	1	838	83.833060	-5.438027	21.387		
	1	810	83.853901	-5.428605	22.269	1	915	83.792900	-5.426330	22.277		
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 4 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o4c	3 Shutter Slitlet	83.829732916666 66 Degrees - 5.4255919444444 2 Degrees	218.71497028904 568			3	18	4058.499
	2	1 (PRISM/CLEAR)	c1 : o4d	3 Shutter Slitlet	83.827889583333 33 Degrees - 5.4285580555555 27 Degrees	218.71514830079 263			3	18	4058.499
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 218.7162 to 218.7162 Degrees (V3 80.14168 to 80.14168) Sequence Observations 4, 41, Non-interruptible										

Proposal 7782 - Observation 5 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 5: o5a, o5b										
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Coordinated Parallel Template(s): NIRCcam Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Data Excess over lower threshold										
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 5:1) Warning (Form): The recommended value is 8 Reference Stars for this template.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	o5	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 7 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging					
	TA Method: MSATA					Module: ALL					
	HFF Readout Mode: false					Subarray: FULL					
	Obtain Confirmation Images: No										
	Science Aperture: MSA Center										
	Primary Candidate List: p5 (385 sources)										
	Filler Candidate List: f5 (150 sources)										
	Spectral Overlap Map: jwst-nirspec-prism										
Spectral Overlap Threshold: 1.5											
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	183	83.745404	-5.363429	22.287	1	946	83.758288	-5.370862	21.187	
	1	493	83.756498	-5.408433	20.975	1	981	83.730687	-5.366411	20.886	
	1	685	83.730512	-5.362036	20.603	1	1002	83.720007	-5.398413	21.128	
	1	933	83.771575	-5.374047	22.497						
Dithers	#	Dither Type									
	1	NONE									

Proposal 7782 - Observation 5 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o5b	3 Shutter Slitlet	83.742008333333 33 Degrees - 5.385128333333 41 Degrees	218.73556026981 436			3	18	4058.499
	2	1 (PRISM/CLEAR)	c1 : o5a	3 Shutter Slitlet	83.737354166666 66 Degrees - 5.383367222222 05 Degrees	218.73599279923 596			3	18	4058.499
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	2	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 218.7287 to 218.7287 Degrees (V3 80.15414 to 80.15414) Sequence Observations 5, 51, Non-interruptible										

Proposal 7782 - Observation 6 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 6: o6a, o6b, o6c, o6d, o6e											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(o6a, o6b, o6c, o6d, o6e (Obs 6)) Warning (Form): Config c1 : o6a (#1) has 1 primary slits affected by failed closed shutters.											
	(o6a, o6b, o6c, o6d, o6e (Obs 6)) Warning (Form): Config c1 : o6b (#3) has 1 master background shutters affected by failed open or closed shutters.											
	(o6a, o6b, o6c, o6d, o6e (Obs 6)) Warning (Form): Config c1 : o6c (#4) has 1 primary slits affected by failed closed shutters.											
	(o6a, o6b, o6c, o6d, o6e (Obs 6)) Warning (Form): Config c1 : o6d (#5) has 1 master background shutters affected by failed open or closed shutters.											
	(o6a, o6b, o6c, o6d, o6e (Obs 6)) Warning (Form): Config c1 : o6d (#5) has 1 primary slits affected by failed closed shutters.											
	(Visit 6:1) Warning (Form): Data Excess over lower threshold											
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	(Visit 6:1) Warning (Form): The recommended value is 8 Reference Stars for this template.											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	o6	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F110W; Readout: NRSRAPID; 7 sources in 4 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p6 (426 sources)											
	Filler Candidate List: f6 (109 sources)											
Spectral Overlap Map: jwst-nirspec-hr												
Spectral Overlap Threshold: 1.5												
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	59	83.819942	-5.437322	20.818	1	132	83.796443	-5.379230	21.653		
	1	65	83.794462	-5.414853	20.410	1	515	83.800062	-5.384670	20.056		
	1	110	83.800850	-5.377228	20.352	1	859	83.820776	-5.408471	20.307		
	1	113	83.790980	-5.393219	21.594							

Proposal 7782 - Observation 6 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Dithers	#	Dither Type									
	1	NONE									
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o6a	3 Shutter Slitlet	83.803450416666 66 Degrees - 5.4139366666666 43 Degrees	216.99949435045 573			3	18	4058.499
	2	1 (PRISM/CLEAR)	c1 : o6e	3 Shutter Slitlet	83.807497083333 33 Degrees - 5.4155794444444 61 Degrees	216.99911918973 59			3	18	4058.499
	3	1 (PRISM/CLEAR)	c1 : o6b	3 Shutter Slitlet	83.799446666666 67 Degrees - 5.4179105555555 79 Degrees	216.99987645678 58			3	18	4058.499
	4	2 (PRISM/CLEAR)	c1 : o6c	3 Shutter Slitlet	83.795192083333 33 Degrees - 5.4138525000000 19 Degrees	217.00026648165 735			3	6	1352.833
	5	2 (PRISM/CLEAR)	c1 : o6d	3 Shutter Slitlet	83.799674583333 33 Degrees - 5.4259591666666 44 Degrees	216.99987060312 42			3	6	1352.833
	Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1		F182M	F360M	SHALLOW2	3	2	6	3	805.258		
2		F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
3		F182M	F360M	SHALLOW2	3	2	6	3	805.258		
4		F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
5		F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements		No Parallel Attachments									
	MSA Scheduled Aperture PA 216.9983 to 216.9983 Degrees (V3 78.42377 to 78.42377)										
	Sequence Observations 6, 61, Non-interruptible										

Proposal 7782 - Observation 7 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 7: o7a, o7b, o7c, o7d, o7e											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(o7a, o7b, o7c, o7d, o7e (Obs 7)) Warning (Form): Config c1 : o7a (#5) has 1 primary slits affected by failed closed shutters.											
	(o7a, o7b, o7c, o7d, o7e (Obs 7)) Warning (Form): Config c1 : o7b (#4) has 1 primary slits affected by failed closed shutters.											
	(Visit 7:1) Warning (Form): Data Excess over lower threshold											
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 7:1) Warning (Form): The recommended value is 8 Reference Stars for this template.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(9)	o7	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F110W; Readout: NRSRAPID; 7 sources in 4 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p7 (353 sources)											
	Filler Candidate List: f7 (182 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Spectral Overlap Threshold: 1.5												
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	53	83.831968	-5.371722	21.316	1	499	83.847259	-5.346673	21.252		
	1	60	83.811921	-5.333919	21.292	1	509	83.821114	-5.326022	20.362		
	1	130	83.813939	-5.362221	21.521	1	513	83.815390	-5.330094	21.512		
	1	177	83.809051	-5.335546	21.910							
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 7 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o7c	3 Shutter Slitlet	83.825824583333 33 Degrees - 5.356900555555 69 Degrees	217.04000746353 395			3	15	3382.083
	2	2 (PRISM/CLEAR)	c1 : o7e	3 Shutter Slitlet	83.83494125 Degrees - 5.362357777777 88 Degrees	217.03916772603 043			3	12	2705.666
	3	2 (PRISM/CLEAR)	c1 : o7d	3 Shutter Slitlet	83.837413333333 33 Degrees - 5.362144444444 25 Degrees	217.03893649090 73			3	12	2705.666
	4	1 (PRISM/CLEAR)	c1 : o7b	3 Shutter Slitlet	83.834345 Degrees - 5.355528055555 67 Degrees	217.03920896312 283			3	15	3382.083
	5	1 (PRISM/CLEAR)	c1 : o7a	3 Shutter Slitlet	83.837400833333 33 Degrees - 5.351842222222 17 Degrees	217.03891584369 75			3	15	3382.083
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	4	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	5	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments										
	MSA Scheduled Aperture PA 217.0411 to 217.0411 Degrees (V3 78.46649 to 78.46649)										

Proposal 7782 - Observation 8 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Observation	Proposal 7782, Observation 8: o8a, o8b, o8c, o8d											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(o8a, o8b, o8c, o8d (Obs 8)) Warning (Form): Config c1 : o8b (#2) has 1 primary slits affected by failed closed shutters.											
	(o8a, o8b, o8c, o8d (Obs 8)) Warning (Form): Config c1 : o8d (#1) has 1 primary slits affected by failed closed shutters.											
	(Visit 8:1) Warning (Form): Data Excess over lower threshold											
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 8:1) Warning (Form): The recommended value is 8 Reference Stars for this template.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(8)	o8	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 7 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy					NIRCам Imaging						
	TA Method: MSATA					Module: ALL						
	HFF Readout Mode: false					Subarray: FULL						
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p8 (378 sources)											
	Filler Candidate List: f8 (157 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Reference Stars	Spectral Overlap Threshold: 1.5											
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	41	83.772615	-5.350559	22.140	1	931	83.770727	-5.365519	22.999		
	1	182	83.764611	-5.346259	21.783	1	938	83.765346	-5.368196	21.313		
	1	287	83.779438	-5.385142	20.804	1	1065	83.765877	-5.346632	22.519		
	1	519	83.773786	-5.379574	20.978							
Dithers	#	Dither Type										
	1	NONE										

Proposal 7782 - Observation 8 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Objects

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o8d	3 Shutter Slitlet	83.779915833333 33 Degrees - 5.364402777777 98 Degrees	216.97235441228 932			3	15	3382.083
	2	1 (PRISM/CLEAR)	c1 : o8b	3 Shutter Slitlet	83.776720833333 33 Degrees - 5.363016666666 81 Degrees	216.97265035256 265			3	15	3382.083
	3	2 (PRISM/CLEAR)	c1 : o8c	3 Shutter Slitlet	83.773018333333 33 Degrees - 5.355438888888 98 Degrees	216.97298044385 778			3	6	1030.73
	4	1 (PRISM/CLEAR)	c1 : o8a	3 Shutter Slitlet	83.784138333333 33 Degrees - 5.357154444444 61 Degrees	216.97194391385 95			3	15	3382.083
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	4	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Scheduled Aperture PA 216.9691 to 216.9691 Degrees (V3 78.39453 to 78.39453) Sequence Observations 8, 81, Non-interruptible										

Proposal 7782 - Observation 16 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 16: o6a, o6b, o6c, o6d, o6e										Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Coordinated Parallel Template(s): NIRCcam Imaging										
Comments: Repeat of faile observation 6											
Diagnostics	(o6a, o6b, o6c, o6d, o6e (Obs 16)) Warning (Form): Config c1 : o6a (#1) has 1 primary slits affected by failed closed shutters.										
	(o6a, o6b, o6c, o6d, o6e (Obs 16)) Warning (Form): Config c1 : o6b (#3) has 1 master background shutters affected by failed open or closed shutters.										
	(o6a, o6b, o6c, o6d, o6e (Obs 16)) Warning (Form): Config c1 : o6c (#4) has 1 primary slits affected by failed closed shutters.										
	(o6a, o6b, o6c, o6d, o6e (Obs 16)) Warning (Form): Config c1 : o6d (#5) has 1 master background shutters affected by failed open or closed shutters.										
	(o6a, o6b, o6c, o6d, o6e (Obs 16)) Warning (Form): Config c1 : o6d (#5) has 1 primary slits affected by failed closed shutters.										
	(Visit 16:1) Warning (Form): Data Excess over lower threshold										
(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(5)	o6	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000								
Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	NIRSpec MultiObject Spectroscopy					NIRCcam Imaging					
	TA Method: MSATA					Module: ALL					
	HFF Readout Mode: false					Subarray: FULL					
	Obtain Confirmation Images: No										
	Science Aperture: MSA Center										
	Primary Candidate List: p6 (426 sources)										
	Filler Candidate List: f6 (109 sources)										
	Spectral Overlap Map: jwst-nirspec-hr										
Spectral Overlap Threshold: 1.5											
Reference Stars											
Dithers	#	Dither Type									
	1	NONE									

Proposal 7782 - Observation 16 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o6a	3 Shutter Slitlet	83.803450416666 66 Degrees - 5.413936666666 43 Degrees	216.99945465042 893			3	18	4058.499
	2	1 (PRISM/CLEAR)	c1 : o6e	3 Shutter Slitlet	83.807497083333 33 Degrees - 5.415579444444 61 Degrees	216.99907948971 284			3	18	4058.499
	3	1 (PRISM/CLEAR)	c1 : o6b	3 Shutter Slitlet	83.799446666666 67 Degrees - 5.417910555555 79 Degrees	216.99983675675 233			3	18	4058.499
	4	2 (PRISM/CLEAR)	c1 : o6c	3 Shutter Slitlet	83.795192083333 33 Degrees - 5.413852500000 19 Degrees	217.00022678162 156			3	6	1352.833
	5	2 (PRISM/CLEAR)	c1 : o6d	3 Shutter Slitlet	83.799674583333 33 Degrees - 5.425959166666 44 Degrees	216.99983090308 598			3	6	1352.833
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	2	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	3	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
	4	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	5	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
Special Requirements	No Parallel Attachments MSA Planned Aperture PA 216.9983 to 216.9983 Degrees (V3 78.4237303 to 78.4237303)										
	Sequence Observations 16, 63, Non-interruptible										

Proposal 7782 - Observation 17 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 17: o17a, o17c, o17d2, o17b2											Thu Mar 12 22:01:35 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCam Imaging											
Diagnostics	(o17a, o17c, o17d2, o17b2 (Obs 17)) Warning (Form): Config c1 : o17a (#1) has 1 primary slits affected by failed closed shutters.											
	(o17a, o17c, o17d2, o17b2 (Obs 17)) Warning (Form): Config c1 : o17b2 (#3) has 2 primary slits affected by failed closed shutters.											
	(o17a, o17c, o17d2, o17b2 (Obs 17)) Warning (Form): Config c1 : o17d2 (#2) has 1 filler slits affected by failed closed shutters.											
	(o17a, o17c, o17d2, o17b2 (Obs 17)) Warning (Form): Config c1 : o17d2 (#2) has 2 primary slits affected by failed closed shutters.											
	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 17:1) Warning (Form): The recommended value is 8 Reference Stars for this template.											
	(Visit 17:2) Warning (Form): Data Excess over lower threshold											
	(Visit 17:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(10)	o17	RA: 05 35 15.6480 (83.8152000d) Dec: -05 23 8.17 (-5.38560d) Equinox: J2000									
	Comments:											
	Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F110W; Readout: NRSRAPID; 6 sources in 3 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2	Filter: F110W; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCam Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: p17 (322 sources)											
	Filler Candidate List: f17 (213 sources)											
	Spectral Overlap Map: jwst-nirspec-hr											
Spectral Overlap Threshold: 1.5												

Proposal 7782 - Observation 17 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	52	83.829511	-5.341369	21.438	1	499	83.847259	-5.346673	21.252	
	1	60	83.811921	-5.333919	21.292	1	509	83.821114	-5.326022	20.362	
	1	130	83.813939	-5.362221	21.521	1	513	83.815390	-5.330094	21.512	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	50	83.840079	-5.361087	21.272	2	130	83.813939	-5.362221	21.521	
	2	54	83.827430	-5.353651	20.656	2	499	83.847259	-5.346673	21.252	
	2	63	83.800496	-5.352120	21.115	2	509	83.821114	-5.326022	20.362	
2	100	83.862499	-5.347685	21.218	2	510	83.823873	-5.362438	20.257		
Dithers	#										
	Dither Type										
	1 NONE										
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1 : o17a	3 Shutter Slitlet	83.828936083333 33 Degrees - 5.3597025000000 26 Degrees	217.88144444172 684			3	21	4734.916
	2	2 (PRISM/CLEAR)	c1 : o17d2	3 Shutter Slitlet	83.819563875 Degrees - 5.3511513888888 66 Degrees	217.88230184069 417			3	6	1352.833
	3	1 (PRISM/CLEAR)	c1 : o17b2	3 Shutter Slitlet	83.837755916666 67 Degrees - 5.3515563888888 66 Degrees	217.88060336161 4			3	21	4734.916
	4	1 (PRISM/CLEAR)	c1 : o17c	3 Shutter Slitlet	83.826641083333 33 Degrees - 5.350382222222 08 Degrees	217.88163897466 737			3	21	4734.916
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	2	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	3	F162M+F150W2	F444W	SHALLOW2	3	2	6	3	805.258		
	4	F182M	F360M	SHALLOW2	3	2	6	3	805.258		
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 217.8827818426656 to 217.8827818426656 Degrees (V3 79.3082121426656 to 79.3082121426656) [MSA Selected] Visits Same PA No Parallel Attachments MSA Scheduled Aperture PA 217.8828 to 217.8828 Degrees (V3 79.3082121426656 to 79.3082121426656)										
	Sequence Observations 17, 71, Non-interruptible										

Proposal 7782 - Observation 11 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 11: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 1, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 11)) Warning (Form): PARALLEL requirement expected. (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 12 After 11 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 1, 11, Non-interruptible							

Proposal 7782 - Observation 12 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 12: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	12 After 11 by 12 Hours to 2 Days, Exclusive Use Of Instrument							

Proposal 7782 - Observation 21 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 21: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 2, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 21)) Warning (Form): PARALLEL requirement expected. (Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 22 After 21 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 2, 21, Non-interruptible							

Proposal 7782 - Observation 22 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 22: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark								Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels		
	ALL	FULL	Imaging				4		
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM8	10	1	1	1	1052.203		
Special Requirements	22 After 21 by 12 Hours to 2 Days, Exclusive Use Of Instrument								

Proposal 7782 - Observation 31 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 31: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 3, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 31)) Warning (Form): PARALLEL requirement expected. (Visit 31:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 32 After 31 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 3, 31, Non-interruptible							

Proposal 7782 - Observation 32 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 32: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 32:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	32 After 31 by 12 Hours to 2 Days, Exclusive Use Of Instrument							

Proposal 7782 - Observation 41 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 41: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 4, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 41)) Warning (Form): PARALLEL requirement expected. (Visit 41:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 42 After 41 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 4, 41, Non-interruptible							

Proposal 7782 - Observation 42 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 42: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark								Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 42:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels		
	ALL	FULL	Imaging				4		
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM8	10	1	1	1	1052.203		
Special Requirements	42 After 41 by 12 Hours to 2 Days, Exclusive Use Of Instrument								

Proposal 7782 - Observation 51 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 51: Move NIRCcam PWs to FLAT							Thu Mar 12 22:01:35 GMT 2026						
	Diagnostic Status: Warning													
	Observing Template: NIRCcam Dark													
	Comments: The Persistence Risk SR actually applies to Obs 5, but is used here for persistence-mitigation implementation purposes.													
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 51)) Warning (Form): PARALLEL requirement expected.													
	(Visit 51:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Template	Module		Subarray		Science Template		Occulting Mask		No. of Output Channels					
	ALL		FULL		Imaging				4					
Spectral Elements	#	Readout Pattern		Groups/Int	Integrations/Exp		Exposures/Dith		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	RAPID		2	1		1		1		21.474			
Special Requirements	Persistence Risk													
	52 After 51 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 5, 51, Non-interruptible													

Proposal 7782 - Observation 52 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 52: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 52:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	52 After 51 by 12 Hours to 2 Days, Exclusive Use Of Instrument							

Proposal 7782 - Observation 61 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 61: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 6, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 61)) Warning (Form): PARALLEL requirement expected. (Visit 61:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 62 After 61 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 6, 61, Non-interruptible							

Proposal 7782 - Observation 62 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 62: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 62:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	62 After 61 by 12 Hours to 2 Days, Exclusive Use Of Instrument							

Proposal 7782 - Observation 63 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 63: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 6, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 63)) Warning (Form): PARALLEL requirement expected. (Visit 63:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 64 After 63 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 16, 63, Non-interruptible							

Proposal 7782 - Observation 64 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 64: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark								Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 64:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels		
	ALL	FULL	Imaging				4		
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM8	10	1	1	1	1052.203		
Special Requirements	64 After 63 by 12 Hours to 2 Days, Exclusive Use Of Instrument								

Proposal 7782 - Observation 71 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 71: Move NIRCcam PWs to FLAT							Thu Mar 12 22:01:35 GMT 2026						
	Diagnostic Status: Warning													
	Observing Template: NIRCcam Dark													
	Comments: The Persistence Risk SR actually applies to Obs 7, but is used here for persistence-mitigation implementation purposes.													
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 71)) Warning (Form): PARALLEL requirement expected.													
	(Visit 71:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Template	Module		Subarray		Science Template		Occulting Mask		No. of Output Channels					
	ALL		FULL		Imaging				4					
Spectral Elements	#	Readout Pattern		Groups/Int	Integrations/Exp		Exposures/Dith		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	RAPID		2	1		1		1		21.474			
Special Requirements	Persistence Risk													
	72 After 71 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 17, 71, Non-interruptible													

Proposal 7782 - Observation 72 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 72: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 72:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL	Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM8	10	1	1	1	1052.203	
Special Requirements	72 After 71 by 12 Hours to 2 Days, Exclusive Use Of Instrument							

Proposal 7782 - Observation 81 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 81: Move NIRCcam PWs to FLAT Diagnostic Status: Warning Observing Template: NIRCcam Dark Comments: The Persistence Risk SR actually applies to Obs 8, but is used here for persistence-mitigation implementation purposes.							Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Move NIRCcam PWs to FLAT (Obs 81)) Warning (Form): PARALLEL requirement expected. (Visit 81:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Template	Module	Subarray	Science Template	Occulting Mask	No. of Output Channels			
	ALL	FULL	Imaging		4			
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
	1	RAPID	2	1	1	1	21.474	
Special Requirements	Persistence Risk 82 After 81 by 12 Hours to 2 Days, Exclusive Use Of Instrument Sequence Observations 8, 81, Non-interruptible							

Proposal 7782 - Observation 82 - A Spectroscopic Survey of the Orion Nebula Cluster: Protostars, Proplyds, and Planetary Mass Obje...

Observation	Proposal 7782, Observation 82: Residual Persistence Check Diagnostic Status: Warning Observing Template: NIRCam Dark								Thu Mar 12 22:01:35 GMT 2026
Diagnostics	(Visit 82:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Template	Module	Subarray		Science Template		Occulting Mask		No. of Output Channels	
	ALL	FULL		Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM8	10	1	1	1	1052.203		
Special Requirements	82 After 81 by 12 Hours to 2 Days, Exclusive Use Of Instrument								